Andhra University College of Engineering

Civil Engineering | Course Code: CE301

Structural Analysis - Semester Exam Question Bank & Worked Solutions

---

Syllabus Focus: CE301 High-Yield Semester Exam Question Bank & Model Solutions

Summary: Comprehensive question bank with previous semester university examination questions, m

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Energy Theorems & Deflections

" Syllabus Scope: Castigliano first and second theorems, Unit load method, Strain energy in bending and t

Unit 2: Slope Deflection Method

" Syllabus Scope: Derivation of slope deflection equations, Analysis of continuous beams with support set

CE301: Structural Analysis - Advanced Topics & Assessment

Academic Track: Civil Engineering - Units 3, 4, 5 & Examination Matrix

---

II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Moment Distribution Method

" Stiffness and carry-over factors, Distribution factors, Analysis of continuous beams and portal frames witho

Unit 4: Moving Loads & Influence Line Diagrams

" Influence lines for reaction, shear force, and bending moment in simply supported girders, Muller-Breslau pr

Unit 5: Arches & Cable Structures

" Three-hinged and two-hinged circular and parabolic arches, Normal thrust, Radial shear, Bending moment diagr

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Analyze a continuous two-span beam with fixed ends subjected to uniformly distributed loads using Mome
2. Apply Castigliano second theorem to calculate vertical deflection at the loaded node of an indetermina
3. Draw the influence line diagram for bending moment at the center of a 20-meter span girder and find ma

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* C.S. Reddy - Basic Structural Analysis Tata McGraw-Hill 3rd Ed
* R.C. Hibbeler - Structural Analysis Pearson 10th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students